

CS5998 Capstone Project
Submission 01 — Project Definition Document
Measuring and Reducing the Lab-to-Field Accuracy Gap in Plant Leaf Disease
Classification
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1. Problem Statement

Plant leaf disease classifiers trained on the PlantVillage dataset routinely report accuracies above 99%. Mohanty, Hughes and Salathé (2016) reached 99.35% on a held-out PlantVillage test set, but the same model scored 31.4% on images collected from trusted online sources and photographed under conditions unlike the training data. The authors concluded that a more diverse set of training data was needed. The cause is visible in the images themselves: PlantVillage shows a single detached leaf on a plain background under even lighting, whereas a photograph taken in a field contains soil, overlapping leaves, shadows and blur. Published headline accuracies therefore say very little about whether such a model would work for a farmer holding a phone. This project measures that lab-to-field gap on a fixed set of tomato and potato disease classes, and evaluates which training decisions close the most of it.

The problem is defined as multi-class image classification evaluated across two domains: a model is trained on laboratory images and scored on both a held-out laboratory test split and a separate field dataset, with the difference between those two scores treated as the quantity of interest. Four interventions commonly claimed to improve cross-domain performance are then compared: ImageNet pre-training, image augmentation, unfreezing the backbone during fine-tuning, and fine-tuning on a small number of field images. This bounded framing, using image data, deep learning and computer vision techniques, and an end-to-end training and evaluation pipeline with a small inference application, keeps the scope realistic for a single semester while addressing a practical deployment question rather than a leaderboard one.

2. Data Source(s)

Primary training source	PlantVillage (Hughes and Salathé, 2015), 54,306 images across 38 classes and 14 crops. Detached leaves photographed on a uniform background under controlled lighting. Supplies the training, validation and in-domain test splits.
Out-of-domain test source	PlantDoc (Singh et al., 2020), 2,598 images across 13 plant species and up to 17 disease classes, gathered from the internet and hand-annotated over roughly 300 human hours. Real field conditions: cluttered backgrounds, variable light and focus. Held back as the field test set, with a small slice reserved for the field fine-tuning experiment.
Class set	Restricted to tomato and potato, the two crops with the most disease classes present in both datasets. A desk comparison of the published class lists indicates ten to twelve usable overlapping classes, covering healthy leaves plus early blight, late blight, Septoria leaf spot, bacterial spot, leaf mould, mosaic virus, yellow leaf curl virus and spider mite damage. The exact set will be fixed after

	counting images per class in both datasets, and recorded in the repository.
Availability and provenance	Both datasets are public and free for research use. Neither is committed to the repository; a download script and a split script will reproduce the data preparation without redistributing images. Dataset sizes quoted above are the published figures and will be replaced by my own counts after the audit.

An initial reading of the two sources identifies a limitation that shapes the design. PlantDoc labels are noisier than PlantVillage labels because the images were scraped from the internet rather than photographed to a protocol, so a sample of each class will be hand-checked and any exclusions documented. PlantDoc is also small, which means most of it must be held back as an untouched test set and only a small slice used for fine-tuning. No image will appear in both a training split and a test split.

3. Intended Methods

- **Deep Learning and Computer Vision (CS5802 Advanced AI):** an ImageNet pre-trained EfficientNet-B0 as the improved model, first as a frozen feature extractor with a new classification head and then fully fine-tuned, with ResNet-50 as a second architecture. Grad-CAM overlays to check whether the model attends to the lesion or to the background, which is a testable explanation for domain sensitivity rather than an assumed one.
- **Machine Learning (CS5617 Data Science):** a colour-histogram and linear SVM baseline to establish how much of the task is solvable without deep learning, and a small CNN trained from scratch as a second baseline to isolate the contribution of pre-training. Metric choice under class imbalance and calibration of the final model.
- **Data Engineering and Preprocessing (CS5617 Data Science):** dataset audit and class mapping across two independently produced sources, stratified and seeded split design, a cached preprocessing pipeline, and an augmentation strategy targeted at the differences between the domains (random resized cropping, colour jitter, rotation, and random erasing to imitate occlusion).
- **Model Evaluation:** every model scored on both test sets, reporting accuracy, macro F1 and the gap between domains, with per-class confusion matrices on the field set. Four ablations, each changing a single key in a configuration file: pre-trained against from scratch, augmentation on against off, frozen against unfrozen backbone, and input resolution or model size. Headline runs repeated with three random seeds and reported as mean and standard deviation, so that differences are distinguished from run-to-run noise.

No external LLM API is used in this project. Training runs on an Apple M5 MacBook Pro (16 GB unified memory) using PyTorch with the Metal backend, with Colab or Kaggle as a fallback and as an independent reproduction check.

4. Expected Outputs

- A results table giving, for every stage from baseline to improved model, accuracy and macro F1 on both the laboratory and field test sets with the gap between them, averaged over three seeds.
- Four ablation tables quantifying how much of the gap each intervention closes, and per-class confusion matrices identifying which diseases survive the domain change and which collapse.
- Grad-CAM overlays comparing what the baseline and the best model attend to on the same field images.
- A public GitHub repository containing the full pipeline (download, splits, preprocessing, training, evaluation, interpretability) with configuration files, fixed seeds, logged runs, pinned dependencies and a reproducibility checklist, developed through an incremental commit history.
- A lightweight application that takes an uploaded leaf image and returns the predicted class with its Grad-CAM overlay, used for the technical checkpoint demonstration.

5. Key Risks and Assumptions

- **Class overlap risk:** the project assumes enough tomato and potato classes overlap between the two datasets to support a ten to twelve class problem. The overlap will be audited before any model is built; if it is smaller or noisier than expected, the class count is reduced and the decision documented rather than the datasets changed, with potato alone as a viable fallback.
- **Label quality in the field set:** PlantDoc labels are assumed usable after a hand-checked sample. Because the images were scraped rather than collected to a protocol, some proportion is expected to be wrong, and excluded images will be listed rather than silently dropped.
- **Effect size assumption:** the project assumes PlantDoc differs enough from PlantVillage to produce a measurable gap, which the 31.4% result of Mohanty et al. and visual inspection of both datasets support. Should the gap prove small, the comparison of interventions and the per-class analysis remain reportable findings in their own right.
- **Compute and scope:** an EfficientNet-B0 at 224 pixels is assumed to train in well under an hour per run on the available hardware, making roughly twenty runs feasible. Cached tensors and per-epoch checkpointing mitigate slower training. The class set and model list are fixed by this document, and any further crops or architectures are recorded as future work rather than added mid-project.
- **Reporting and defensibility:** report chapters are drafted from the ninth week and all figures generated by scripts so they can be regenerated if a number changes, and a dated decision log is maintained in the repository as decisions are made rather than reconstructed afterwards.